

Consumer Lens — Technical Architecture & Deployment Documentation

Smart India Hackathon 2024 · Problem Statement ID: 26034
Automated Compliance Verification of Packaged Commodities under Legal Metrology Rules, 2011

Ministry of Consumer Affairs, Food & Public Distribution

Department of Consumer Affairs (DoCA) · Legal Metrology Division

Problem Statement ID: 26034 · Automated Compliance Verification of Packaged Commodities under Legal Metrology (Packaged Commodities) Rules, 2011

1. Executive Summary & Objective

Consumer Lens is an institutional-grade, AI-powered regulatory enforcement platform engineered for Legal Metrology officers, field inspectors, and supervisory authorities across India.

The platform automates the inspection of physical packaged commodities and e-commerce product listings to ensure strict adherence to the **Legal Metrology Act, 2009** and the **Legal Metrology (Packaged Commodities) Rules, 2011 (LMPC Rules, 2011)**.

Key Capabilities

- Multi-Modal Label Evidence Capture:** Live camera ingestion, high-resolution packaging photo uploads, and automated PDP web scraping for major e-commerce portals (Amazon, Flipkart).
- Vision AI Multimodal Extraction:** Gemini Vision multimodal inference detecting normalized 2D bounding boxes (`box_2d: [ymin, xmin, ymax, xmax]`), font size compliance, and print readability.
- Rule 10 & 11 Font Size Compliance:** Evaluates numeral height against packaging net capacity tables, aspect ratio limits, and prominence of MRP.
- Rule 9 & 14 Packaging Readability & Defect Assessment:** Real-time evaluation of contrast adequacy, glare, blur, and text legibility.
- Misleading & Deceptive Declaration Flags:** Automatic detection of unauthorized price sticker overlays, contradictory claims, and non-standard pack sizes.
- Statutory Enforcement Dossier & PDF Memorandum:** Generates courtroom-ready digital inspection dossiers compliant with Section 65B of the Indian Evidence Act, complete with SHA-256 evidence hashes, verification QR codes, and institutional seals.
- Zero-Data-Loss Offline Resilience:** Client-side IndexedDB persistence ensures field inspectors never lose in-progress scans upon browser reloads or network dropouts.

2. High-Level System Architecture

The following diagram illustrates the end-to-end architecture of Consumer Lens, connecting the field inspector interface with the application server, AI vision inference engine, relational persistence layer, and external verification infrastructure.

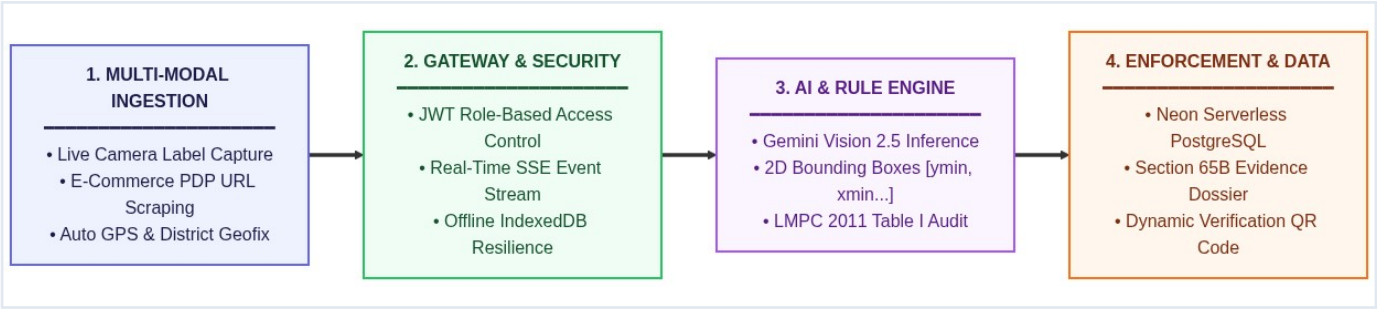


Figure 1: End-to-End System Architecture (4-Tier Ingestion to Enforcement)

3. System Design & User Interaction Flow

The following System Design cum User Flow diagram illustrates the exact synchronization between field inspector actions on the client interface and the asynchronous cloud services executing beneath the surface.

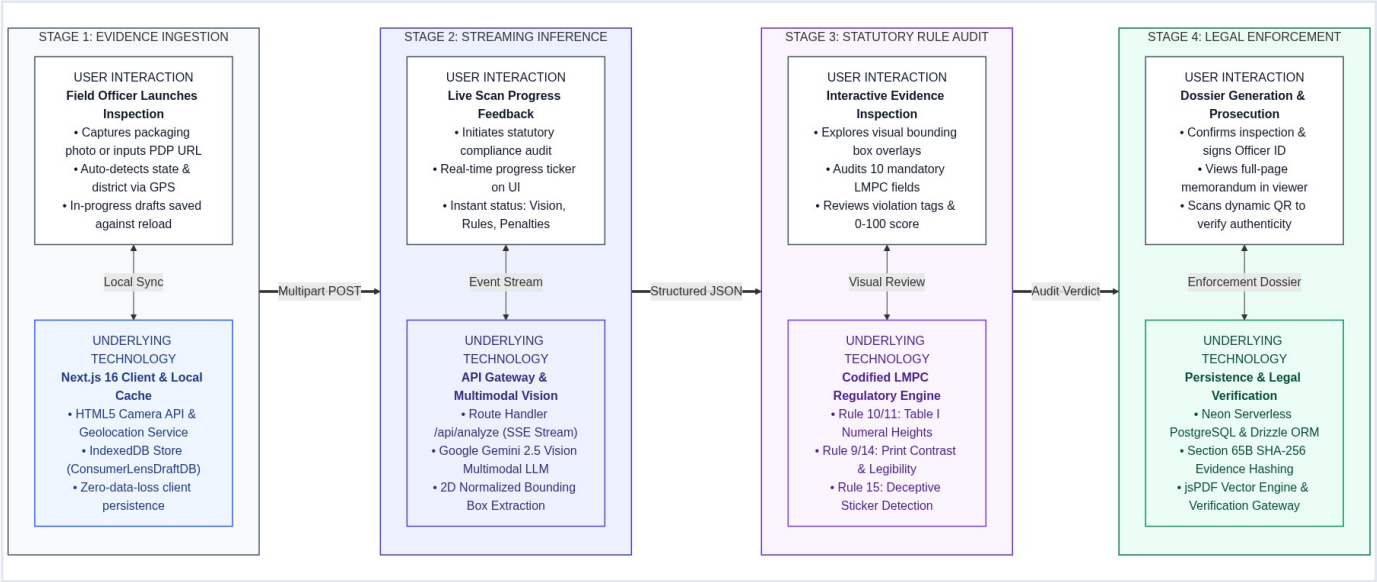


Figure 2: System Design cum User Interaction Flow (Field Officer vs. Cloud Services)

Stage-by-Stage Technical Interaction Matrix

Stage	Field Inspector Interaction	Technology Components Utilized	Data Processing & Protocol
1. Evidence Ingestion	Launches PWA on mobile or workstation; triggers live camera stream or inputs e-commerce PDP URL; state/district detected automatically.	HTML5 MediaDevices API, BigDataCloud Geocoding, client-side IndexedDB (ConsumerLensDraftDB).	Image serialized into Base64; draft state preserved locally to prevent loss on browser reload.
2. Streaming Inference	Clicks "Run Compliance Analysis"; views live status ticker advancing through inference milestones.	Next.js Edge / Node Route Handlers (/api/analyze), Server-Sent Events (text/event-stream), Gemini Vision 2.5 API.	HTTP POST with multipart payload; server dispatches prompt + image to Gemini; streams back real-time step events.
3. Statutory Audit	Navigates visual label inspector; toggles 2D bounding boxes over packaging; audits missing declarations and penalty scores.	React 19 Canvas/SVG coordinate mapper, LMPC Rule 10/11 Table I height engine, Rule 9/14 contrast scorer.	Maps normalized coordinates [ymin, xmin, ymax, xmax] to screen pixels; evaluates declarations against statutory rules; computes 0-100 score.

4. Legal Enforcement	Confirms findings; saves inspection to state repository; launches full-screen PDF dossier; verifies tamper-proof QR code.	Neon Serverless PostgreSQL, Drizzle ORM, Web Crypto SHA-256 hashing, jsPDF vector engine.	Generates cryptographic hash over inspection parameters; stores record in Postgres; issues Section 65B certified PDF memorandum.
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4. LMPC Rules, 2011 Regulatory Decision Engine

The regulatory engine evaluates 10 statutory declaration fields against codified legal requirements from the Legal Metrology (Packaged Commodities) Rules, 2011:

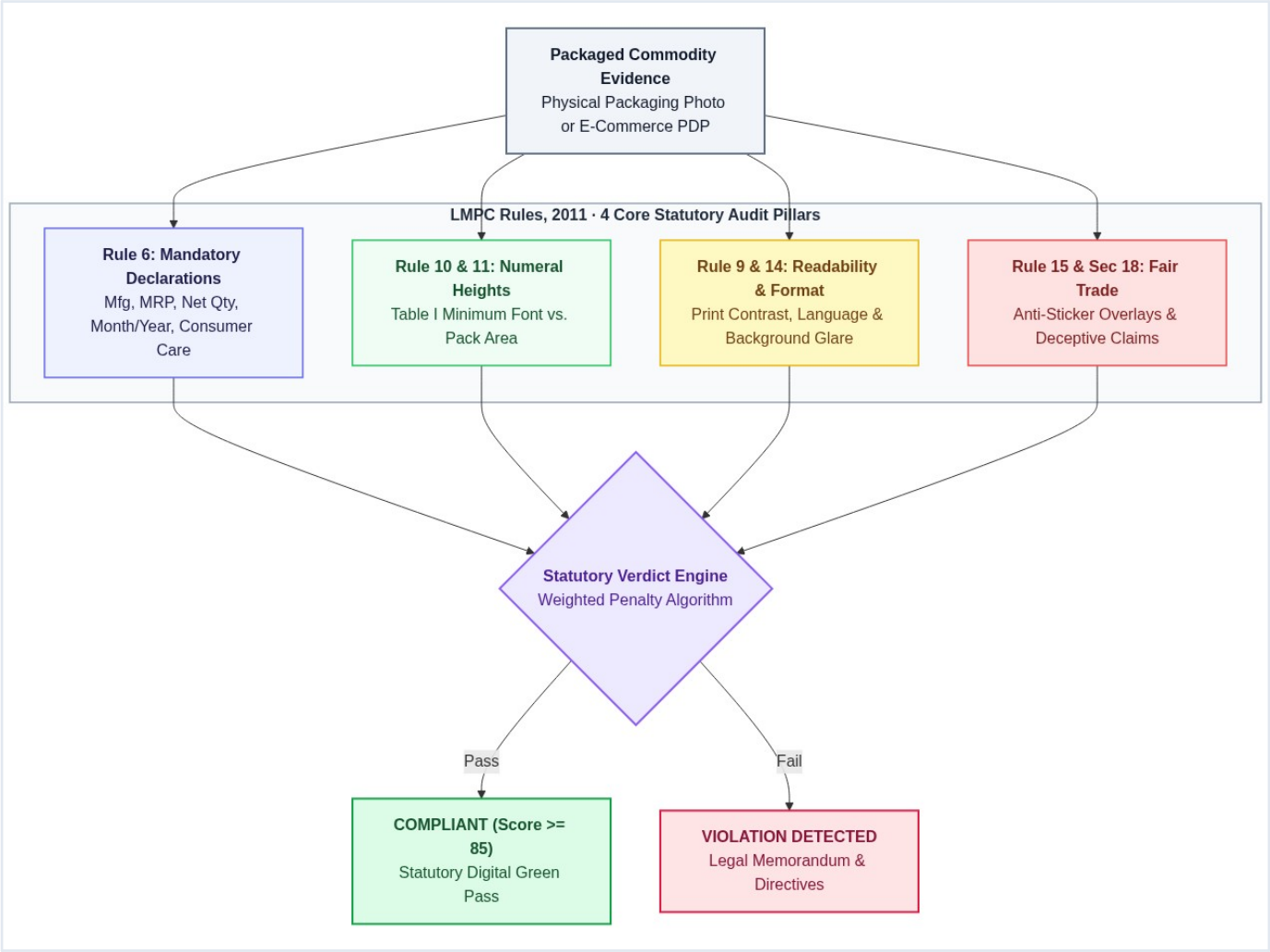


Figure 3: Codified LMPC Rules, 2011 Regulatory Decision Engine

Statutory Severity Matrix

LMPC Rule Citation	Mandatory Declaration	Defect Condition	Severity Rating	Penalty Deduction
Rule 6(1)(a)	Manufacturer / Packer / Importer	Missing name, missing registered address, or no role qualifier	CRITICAL	-25 pts
Rule 6(1)(b)	Generic / Common Name	Brand name only; generic commodity nature omitted	MAJOR	-15 pts
Rule 6(1)(c)	Maximum Retail Price (MRP)	Missing "Inclusive of all taxes", missing currency symbol	CRITICAL	-25 pts

Rule 6(1)(d)	Date of Manufacture / Packing	Missing month and year of packaging	MAJOR	-15 pts
Rule 6(1)(e)	Net Quantity	Non-SI units, missing whitespace around numeral	CRITICAL	-25 pts
Rule 6(2)	Consumer Care Grievance Cell	Missing official phone, email, or physical address	MAJOR	-15 pts
Rule 6(11)	Unit Sale Price (USP)	Missing calculated unit price per gram/ml for packs > 100g/ml	MINOR	-10 pts
Rule 9 & 14	Language & Readability	Low print contrast, glare, blur, or non-English/Hindi declaration	MINOR	-10 pts
Rule 10 & 11	Minimum Font Size (Table I)	Numeral height below statutory minimum for pack area	MAJOR	-15 pts
Rule 15 / Sec 18	Misleading Declarations	Dual pricing stickers, deceptive claims, non-standard pack sizes	CRITICAL	-30 pts

5. Security & Role-Based Access Control (RBAC) Architecture

Consumer Lens implements institutional role segregation aligned with statutory enforcement hierarchies:

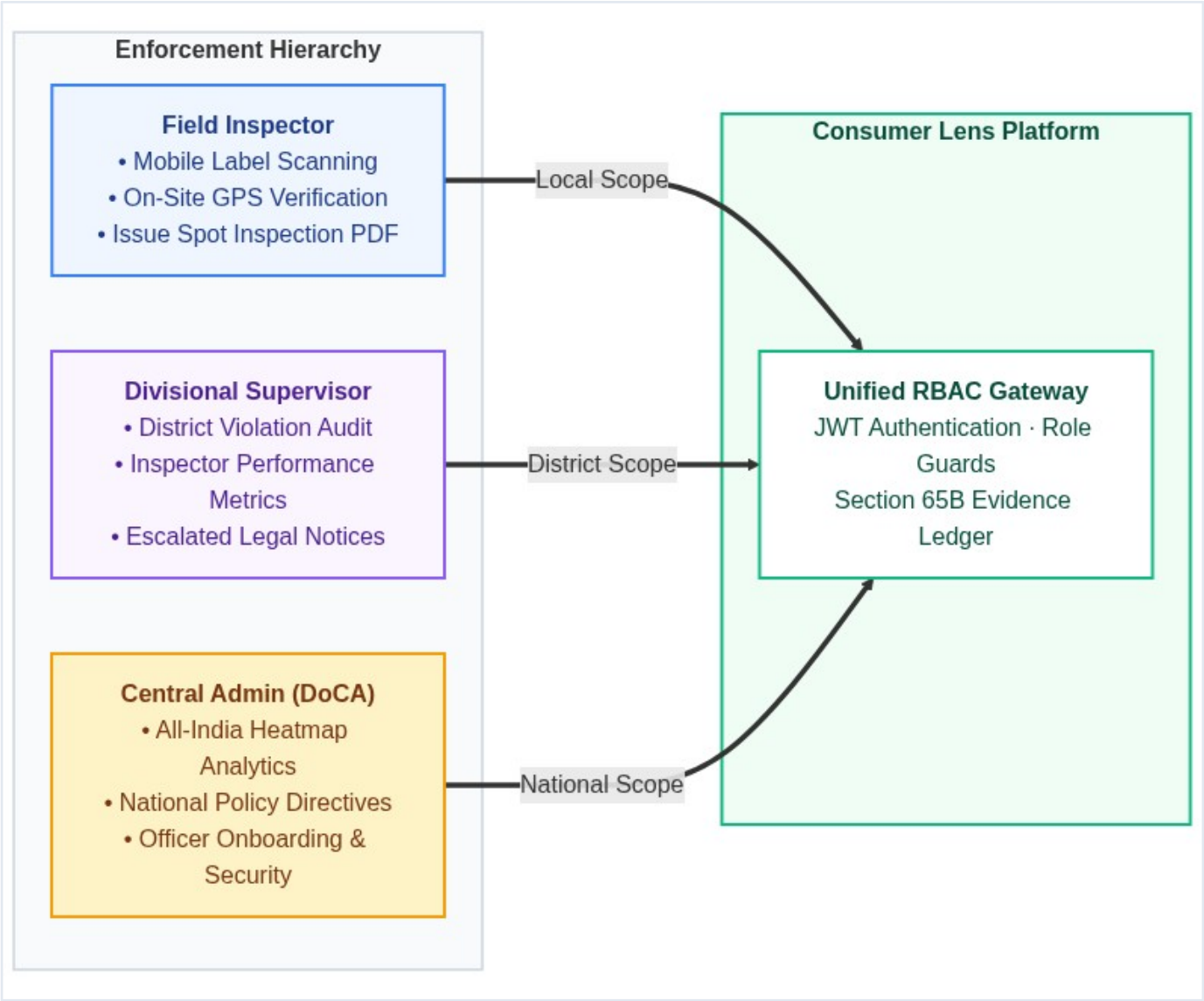


Figure 4: Role-Based Access Control (RBAC) & Enforcement Hierarchy

6. Section 65B Indian Evidence Act Compliance Framework

To ensure that generated inspection reports are legally admissible as electronic evidence in Indian courts of law during prosecution of non-compliant manufacturers, the platform incorporates a statutory digital evidence chain:

1. SHA-256 Cryptographic Evidence Digest:

```
\text{Digest} = \text{SHA-256}(\text{InspectionID} \parallel \text{Date} \parallel \text{Manufacturer} \parallel \text{MRP} \parallel \text{ImageBase64})
```

This cryptographic hash is stamped into the official dossier reference header. Any tampering with the document or evidence image invalidates the hash.

2. Dynamic Verification QR Codes:

Each generated memorandum contains vector QR codes pointing directly to [https://consumer-lens.vercel.app/inspections/\[id\]](https://consumer-lens.vercel.app/inspections/[id]). Enforcement officers, magistrates, and manufacturers can scan the QR code to verify the report's authenticity against the tamper-proof server record.

3. Official Seals & Digital Signatures:

The report renders proportional vector seals:

- National Emblem of India (State Emblem)
- 75 Azadi Ka Amrit Mahotsav Logo
- Food Safety and Standards Authority of India (FSSAI) Seal
- Legal Metrology Directorate Official Inspection Stamp
- Cryptographic vector checkmark and officer employee ID stamp.

7. Production Deployment & Cloud Framework

Consumer Lens is engineered for zero-downtime multi-cloud deployment with containerized support for Google Cloud Run, AWS ECS, or Vercel Edge.

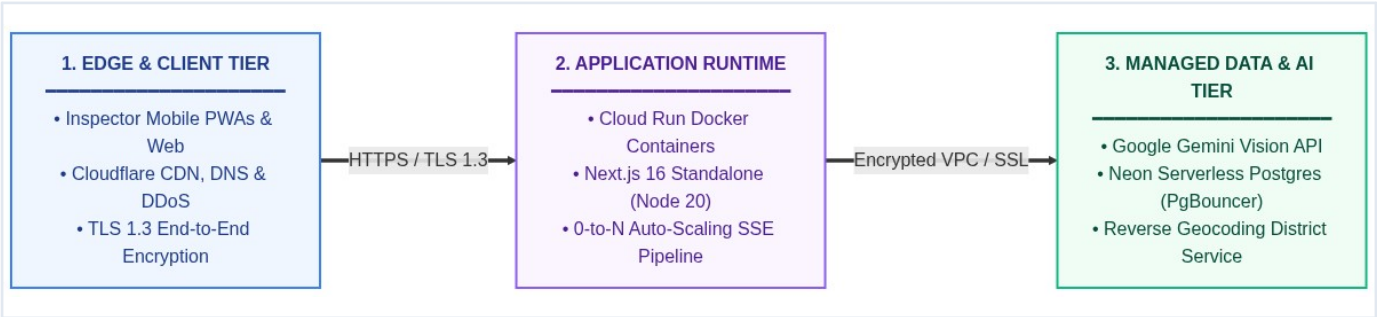


Figure 5: Multi-Cloud High-Availability Deployment Architecture

Multi-Stage Dockerfile Architecture

The application includes a production-hardened multi-stage Docker build optimized for minimal image footprint and Google Cloud Run execution:

```
# Stage 1: Base Image with Node 20 Alpine & Corepack PNPM
FROM node:20-alpine AS base
WORKDIR /app
RUN corepack enable && corepack prepare pnpm@latest --activate

# Stage 2: Dependencies Installation with Cache Mounting
FROM base AS deps
RUN apk add --no-cache libc6-compat
WORKDIR /app
COPY package.json pnpm-lock.yaml* ./
RUN pnpm i --frozen-lockfile

# Stage 3: Builder with Next.js Standalone Optimization
FROM base AS builder
WORKDIR /app
COPY --from=deps /app/node_modules ./node_modules
COPY . .
ENV NEXT_TELEMETRY_DISABLED=1
ENV NODE_ENV=production
RUN pnpm build

# Stage 4: Minimal Runner (Distroless-style Non-Root Execution)
FROM node:20-alpine AS runner
WORKDIR /app
ENV NODE_ENV=production
ENV PORT=8080
ENV HOSTNAME="0.0.0.0"
RUN addgroup --system --gid 1001 nodejs && adduser --system --uid 1001 nextjs
COPY --from=builder /app/public ./public
COPY --from=builder --chown=nextjs:nodejs /app/.next/standalone ./
COPY --from=builder --chown=nextjs:nodejs /app/.next/static ./next/static
USER nextjs
EXPOSE 8080
CMD ["node", "server.js"]
```

8. Relational Schema & Entity-Relationship Model

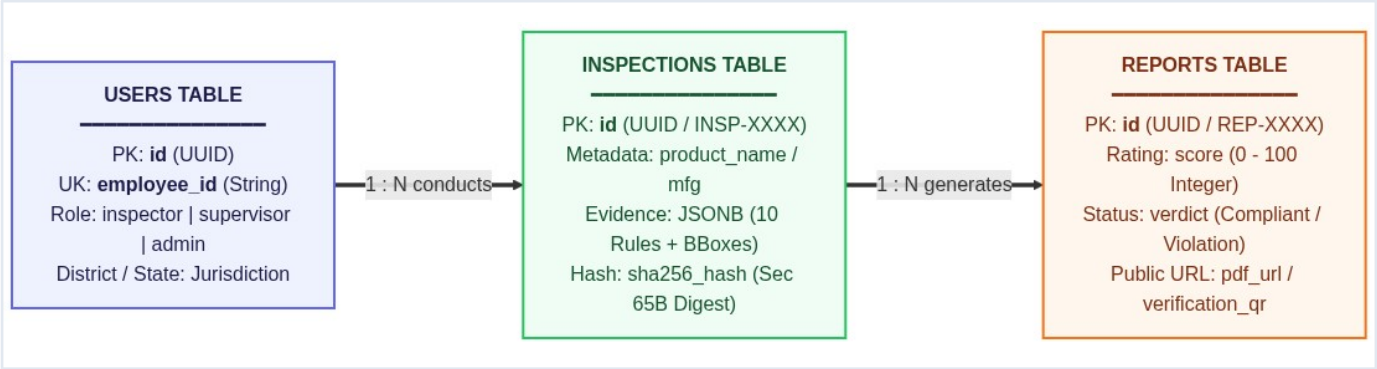


Figure 6: Relational Schema & Entity-Relationship Data Model

9. Presentation Cheat-Sheet (Summary for Evaluation & Slides)

Metric / Dimension	Implementation Detail
Statutory Law	Legal Metrology Act, 2009 & Legal Metrology (Packaged Commodities) Rules, 2011
Target Authority	Department of Consumer Affairs (DoCA), Ministry of Consumer Affairs, Govt. of India
Front-End Stack	Next.js 16.3 (App Router), React 19, Tailwind CSS 4, shadcn/ui, Lucide Icons
AI Inference	Gemini Vision 2.5 Multimodal LLM returning normalized 2D bounding boxes (box_2d)
Font Size Rules	Rule 10 & 11 (Table I minimum numeral height verification against packaging area)
Readability Rules	Rule 9 & 14 (Automated contrast ratio, blur/glare detection, and print quality scoring)
Deceptive Claims	Rule 15 & Section 18 (Price sticker overlay detection & contradictory claim flagging)
Offline Resilience	Native browser IndexedDB (ConsumerLensDraftDB) caching images & scan state
Legal Admissibility	Section 65B Indian Evidence Act SHA-256 evidence digest & dynamic verification QR
Database	PostgreSQL with Drizzle ORM, connection pooling via Neon serverless
Security	JWT HTTP-only cookies, bcrypt hashing, role-based route middleware (Inspector/Supervisor/Admin)
Containerization	Multi-stage Dockerfile running as non-root nextjs user on Alpine Node 20 (Port 8080)